

# Unit 3, Lesson 1: Equivalent Conversions: Fractions $\leftrightarrow$ Decimals; Decimals $\leftrightarrow$ Percentages

## Benchmark

MA.6.NSO.3.5 Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals, and percentages.

## Learning Targets

- ☐ I can convert between fractions and decimals.
- ☐ I can convert between percentages and decimals.

3.1.1 Warm-Up: Self-Evaluation

1. For each problem, rate your level of confidence in your ability to answer each question, but don't actually answer them.

Use a scale from 1 to 5, where 1 means you do not feel confident about how to answer the question, and 5 means you feel very confident.

| Problem                                                | Confidence Level |
|--------------------------------------------------------|------------------|
| Rewrite $2\frac{2}{5}$ in its equivalent decimal form. |                  |
| Rewrite 0.625 as a fraction with the same value.       |                  |
| Rewrite 0.9 as an equivalent percentage.               |                  |
| Rewrite 75% in its equivalent decimal form.            |                  |

3.1.2 Guided Instruction: Fraction and Decimal Conversions

| Decimal Place Value Chart |          |      |      |               |        |            |             |
|---------------------------|----------|------|------|---------------|--------|------------|-------------|
| Thousands                 | Hundreds | Tens | Ones | Decimal Point | Tenths | Hundredths | Thousandths |
|                           |          |      |      |               |        |            |             |



The place value of a number provides the numerical value that a digit has by virtue of its position in a number.

Example: In 278.493  
the 8 is in the **ones** place  
the 4 is in the **tenths** place  
the 9 is in the **hundredths** place  
the 3 is in the **thousandths** place

1. Use the original number to fill in the missing digits and place values.

| Original Number | Digit | Place Value |
|-----------------|-------|-------------|
| 422.54          | 5     | Tenths      |
| 5.276           |       | Ones        |
| 12.749          | 4     |             |
| 3.826           |       | Thousandths |

2. Consider the number 0.08.
- a. In the number 0.08, what is the place value of the final digit?
- b. What is the written description of the fraction  $\frac{8}{100}$ ?
- c. Discuss with your partner any similarities or differences between the decimal 0.08 and the fraction  $\frac{8}{100}$ .



When rewriting a decimal that is greater than 0 but less than 1 as an equivalent fraction, use the place value of the final digit to determine the denominator of the fraction.

3. In the decimal 0.7, the final digit is in the \_\_\_\_\_ place, so \_\_\_\_\_ is the equivalent fraction form.

- ☐  $\frac{7}{10}$
- ☐  $\frac{7}{100}$
- ☐  $\frac{7}{1000}$

- ☐ tenths
- ☐ hundredths
- ☐ thousandths

4. In the decimal 0.037, the final digit is in the \_\_\_\_\_ place, so \_\_\_\_\_ is the equivalent fraction form.

- ☐  $\frac{37}{10}$
- ☐  $\frac{37}{100}$
- ☐  $\frac{37}{1000}$

- ☐ tenths
- ☐ hundredths
- ☐ thousandths

A fraction can be converted into its equivalent decimal form by dividing the **numerator** by the **denominator**.

5. To convert the fraction  $\frac{1}{4}$  to a decimal, divide 1 by 4 by completing the diagram below.

$$4 \overline{) 1.00}$$

**Number of ones  
groups of 4**

—

**Amount that remains**

—

**Number of tenths  
groups of 4**

—

**Amount that remains**

—

**Number of  
hundredths groups  
of 4**

—

**Amount that remains**

—

### 3.1.3 Guided Practice: Fraction and Decimal Conversions

1. Complete the table by converting each decimal to an equivalent fraction. If possible, write the equivalent fraction so that the numerator and denominator have no common factors.

| Decimal Form | Fraction Form  |
|--------------|----------------|
| 0.08         | $\frac{2}{25}$ |
| 0.27         |                |
| 2.2          |                |
| 0.005        |                |

2. Complete the table by rewriting each fraction as its equivalent decimal.

| Fraction Form   | Your Work | Decimal Form |
|-----------------|-----------|--------------|
| $\frac{1}{5}$   |           |              |
| $\frac{5}{8}$   |           |              |
| $\frac{7}{200}$ |           |              |

3.1.4 Guided Instruction: Decimal and Percentage Conversions

A **percentage** is a ratio expressed as a fraction with a denominator of 100.

Percentages can also be written as fractions with a denominator of 100, or as a decimal.

1. Write 66% as a fraction and as a decimal.

| Percent | Fraction            | Decimal |
|---------|---------------------|---------|
| 66%     | $\frac{\quad}{100}$ |         |

To convert a decimal to a percent, multiply the decimal by 100.

2. Convert the decimal 0.16 to a percent.

$$0.16 \times \frac{\quad}{\quad} \%$$

To convert a percent to a decimal, divide the percent by 100 or multiply by  $\frac{1}{100}$ .

3. Convert 16.2% to a decimal.

$$16.2 \div \frac{\quad}{\quad} \text{ or } 16.2 \cdot \frac{\quad}{\quad}$$

### 3.1.5 Guided Practice: Decimal and Percentage Conversions

1. Complete the table by converting each decimal to an equivalent percentage.

| Decimal Form | Percent Form |
|--------------|--------------|
| 0.08         | 8%           |
| 0.27         |              |
| 2.2          |              |
| 0.005        |              |

2. Rewrite each percentage into its equivalent decimal form to complete the table.

| Percent Form | Decimal Form |
|--------------|--------------|
| 72%          | 0.72         |
| 12.5%        |              |
| 0.7%         |              |
| 214%         |              |

**3.1.6 Your Turn!**

1. Rewrite each percentage as a decimal.

a. 125.6%

b. 3%

2. Convert each decimal to an equivalent percentage.

a. 0.008

b. 4.71

3. Rewrite each fraction as a decimal.

a.  $\frac{1}{2}$

b.  $2\frac{1}{10}$

4. Rewrite each decimal as an equivalent fraction. Simplify where applicable.

a. 0.94

b. 1.7

**3.1.7 Wrap-Up: Ticket Out the Door**

1. Circle the expression that does not belong.

|               |               |
|---------------|---------------|
| $\frac{2}{5}$ | 0.4           |
| 40%           | $\frac{4}{8}$ |

2. Explain your reasoning for your choice.